F 1.1

FIZIKALNE KOLIČINE : [ŠT]

"a" - število (skalar)

"$\vec{v}$" - vektor

"$\underline{\underline{A}}$" - tenzorji (matrike, ...)

znanost : SI - SISTEM [ENOTE]
(objektivni sistem)

- čas : atomska ura C (1967)

1 s = 9 192 631 770 prehodov

perioda sevanja v mirujočem stanju

$E = m c_0^2$

- razdalja : definicija svetlobne hitrosti v vakuumu "c_0" ali "c"

$[m/s]$

1 m = pot sv. v $\frac{1}{299\,792\,458}$ s

- masa (gap sesh)

$c_0 = 299\,792\,458$ m/s

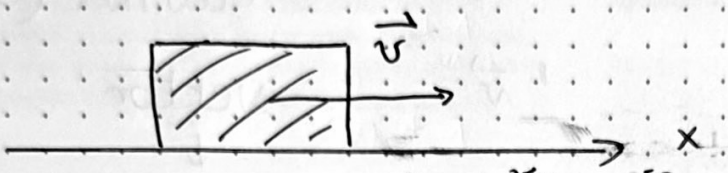
$\approx 3 \cdot 10^8$ m/s

$m_e = 9 \cdot 10^{-31}\ kg$

1 1

ZGLED: 1D GIBANJE (1 os je dovolj)

KINEMATIKA V 1D



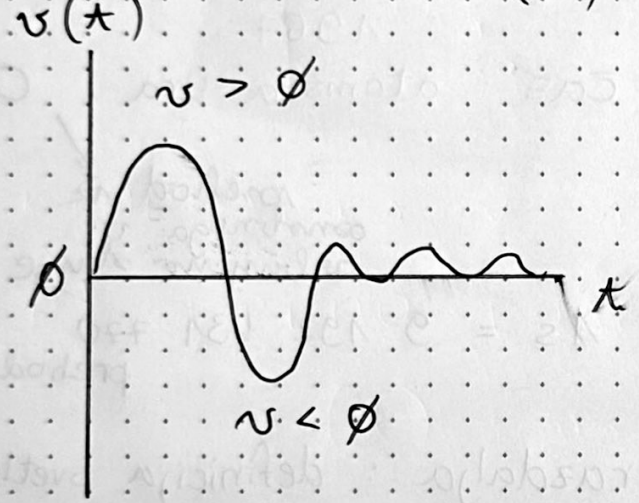
$x: \vec{v} = (v_x, v_y, v_z) = (v, \cancel{0}, \cancel{0})$

$\Uparrow$

$v(t)$

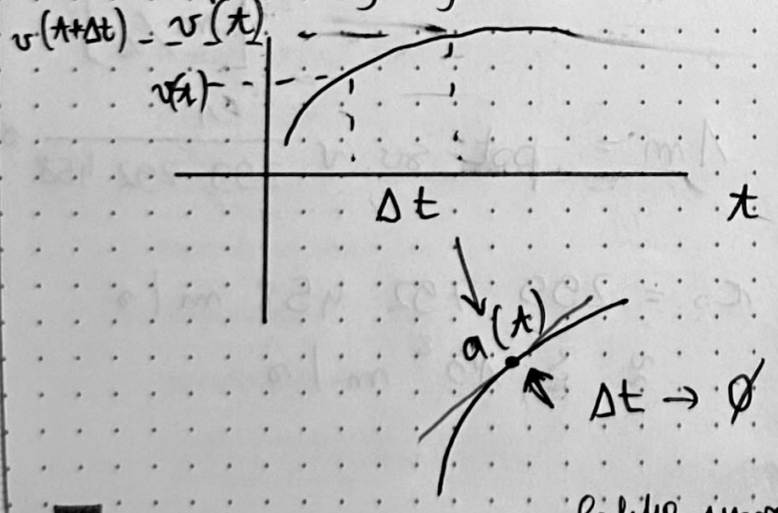
POSPEŠEK $a(t)$

$v(t)$ → LEGA $x(t)$



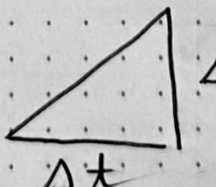
pot: kot kamor pusti lego odmakljon od osi

– spreminjanje hitrosti s časom



$\Delta v = a \cdot \Delta t \Rightarrow a = \frac{\Delta v}{\Delta t}$

$a = \frac{v(t+\Delta t) - v(t)}{\Delta t}$



lahko izpeljalo s Pitagorovim izrekom

$$\Rightarrow a(t) = \lim_{\Delta t \to 0} \frac{v(t+\Delta t) - v(t)}{\Delta t}$$

ODVOD HITROSTI PO ČASU

$$= \frac{dv}{dt} \quad (\text{LEIBNITZ})$$

$$= \dot{v} \quad (\text{NEWTON})$$

tudi: $v(t) = \frac{dx}{dt} = \dot{x}$

ODVOD t^n: (ponovi)

$$y = k \cdot t^n \Rightarrow \dot{y} = k \cdot (n \cdot t^{n-1})$$

npr. $y = k \cdot t \Rightarrow \dot{y} = k \cdot 1 \cdot t^0 = k$

OBRATNA TRANSFORMACIJA (MAT. OPERACIJA) OD ODVODA JE INTEGRACIJA

$$a(t) = \dot{v}(t) = \frac{dv}{dt}$$

$$v(t) = v(t_0) + \int_{t_0}^{t} a(t)\,dt$$

$v(t_0)$ začetna hitrost

$\int_{t_0}^{t} a(t)\,dt$ sprememba

$\Delta v = v(t) - v(t_0)$

npr. $a(t) = B\,t^n$

$$\int_{t_0}^{t} a(t)\,dt = B \cdot \frac{1}{n+1} \cdot t^{n+1} \Big|_{t_0}^{t}$$

$$= B\,\frac{t^{n+1}}{n+1} - B\,\frac{t_0^{n+1}}{n+1}$$

LEGA:

$$x(t) = \underbrace{x(t_0)}_{x_0} + \int v(t)\,dt$$

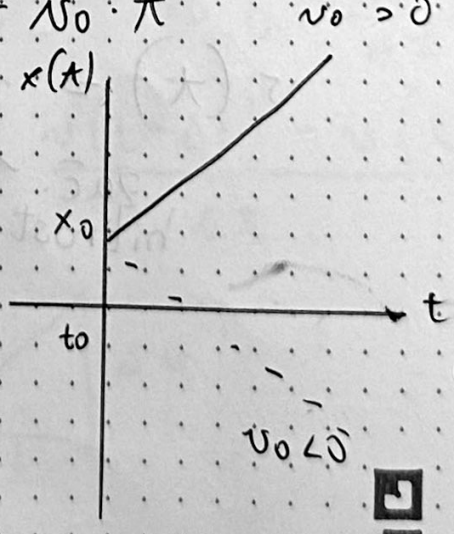
KINEMATIKA V 1D : PREMO GIBANJE

I. PREMO ENAKOMERNO GIBANJE

$v(t) = v_0 \rightarrow a(t) = \dot{v} = \emptyset$

$$\hookrightarrow x(t) = x_0 + \int_{t_0}^{t} v_0\,dt$$

$$= x_0 + v_0 \cdot t$$



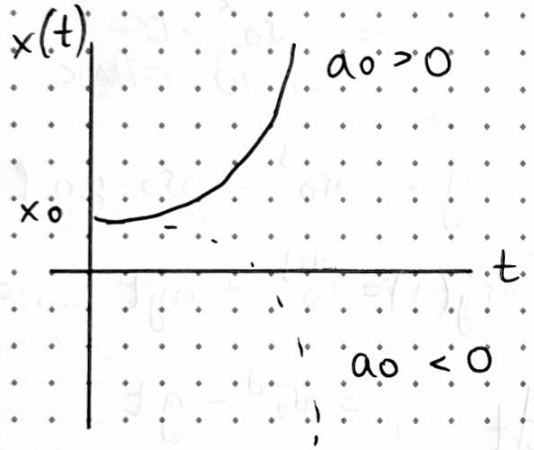
(II) ENAKOMERNO POSPEŠENO GIBANJE

$$a(t) = a_0 \rightarrow v(t) = v_0 + \int_{t_0}^{t} a_0 \cdot dt$$

$$= v_0 + a_0 \cdot t$$

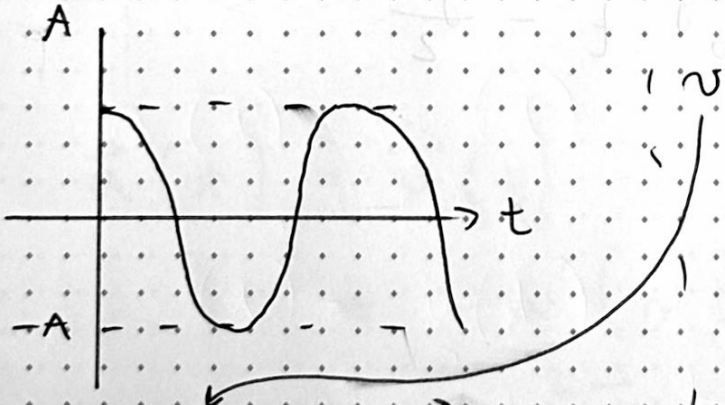
$$x(t) = x_0 + \int_{t_0}^{t} (v_0 + a_0 \cdot t)\, dt$$

$$= x_0 + v_0 \cdot t + \frac{a_0 \cdot t^2}{2}$$



[m]

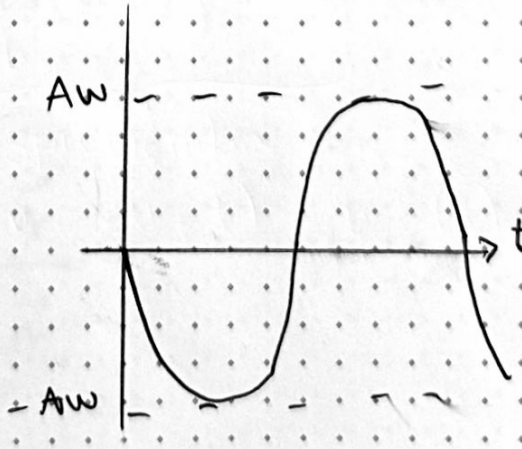
(ZA FORO: NIHANJE) $x(t) = A \cdot \cancel{\sin} \cos(\omega t)$



$$v(t) = \dot{x}(t) = A \cdot (-\sin(\omega t) \cdot \omega)$$

$$= -A\omega \sin(\omega t)$$

[m/s]



$$a(t) = \dot{v}(t) = \ddot{x}(t)$$

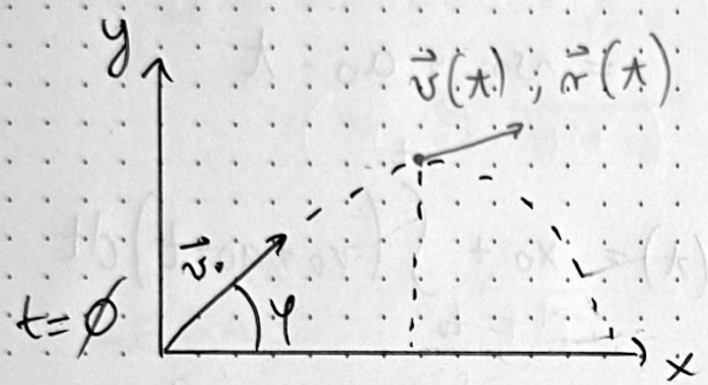
$$= (-A\omega)\cos(\omega t) \cdot \omega$$

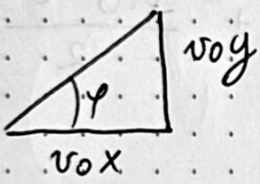
$$= -A\omega^2 \cos(\omega t)$$

$$= -\omega^2 x(t)$$

GIBANJE V PROSTORU

x :
y :
z :
} poračunamo!





POŠEVNI MET : 2 GLED

$x: v_0^x = v_0 \cdot \cos\varphi$

$a_x = 0$

$x(t) = \underbrace{x_0}_{0} + v_0^x \cdot t$

$= v_0^x \cdot t$

PREMO ENAK.

$y: v_0^y = v_0 \cdot \sin\varphi$

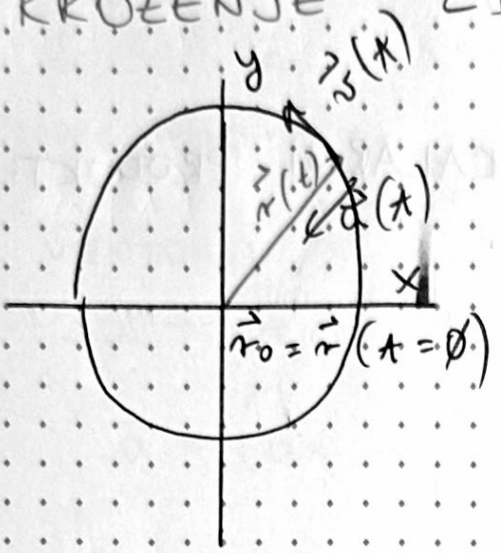
$v_y(t) = v_0^y + a_y t \quad \dots a_y = -g$

$= v_0^y - gt$

$$y(t) = y_0 + \int_{t=0}^{t} (v_0^y - gt)\,dt$$

$$= y_0 + v_0^y \cdot t - \frac{gt^2}{2}$$

KROŽENJE 2D



(+) (−)

$\vec{r}(t) = (x(t), y(t), \emptyset)$

T ... obhodni čas

$\omega = \frac{2\pi}{T} = 2\pi\nu$

ENAKOMERNO KROŽENJE

$$\vec{r}(t) = (r \cdot \cos(\omega t), r \cdot \sin(\omega t), \emptyset)$$

izberemo: $t_0 = \emptyset$: $r(t \ldots) = (r, \emptyset, \emptyset)$

začetni čas

$$\vec{v}(t) = (\dot{x}, \dot{y}, \emptyset) = (\underbrace{r(-\sin(\omega t)\cdot\omega}_{v_x}, \underbrace{r(\cos(\omega t)\cdot\omega}_{v_y}, \emptyset)$$

$$\vec{a}(t) = \dot{\vec{v}}(t)$$

$$= -\omega^2 (r\cos(\omega t), r\sin(\omega t), \emptyset)$$

$$= -\omega^2 \vec{r}$$

LEKCIJE: $\frac{\vec{a} \parallel -\vec{r}}{\vec{v} \perp \vec{r}}$

PREDNOST VEKTORJEV:

$r^2 = \vec{r} \cdot \vec{r}$ ← SKALARNI PRODUKT

ODVAJAJ PO ČASU